

Smart Patio Shield

Live Inference Deployment Guide

This document outlines the deployment details for the Smart Patio Shield inference service. The deployed service utilizes Model 1 (tabular XGBoost) alone, maintaining a lightweight architecture without a satellite pipeline or GPU requirement, as reanalysis weather data provides the necessary predictive value.

1. Repository & Setup

Follow these steps to access and configure the deployment files:

- Navigate to the project GitHub repository: <https://github.com/romayneg/smart-patio-shield>
- Clone or download the repository to your local machine.
- Open the `deploy` folder. Ensure the following core files are present:
 - `smart-patio-shield app.py` : The FastAPI backend service.
 - `dashboard.html` : The interactive frontend console.
 - `xgboost_baseline.json` : The trained XGBoost model file.

2. Install Dependencies

Ensure Python is installed on the host machine. Install the required backend dependencies by executing the following command in the terminal:

```
pip install fastapi uvicorn xgboost pandas numpy
```

3. Running the Service

Step 1: Start the Backend Server

Open a terminal, navigate to the `deploy` folder, and start the FastAPI server:

```
python "smart-patio-shield app.py"
```

The terminal will confirm success with *"Model loaded"* and indicate that Uvicorn is running on `http://127.0.0.1:8000`. Leave this terminal running in the background.

Step 2: Open the Dashboard

Double-click `dashboard.html` from the `deploy` folder to open it in a web browser.

Verify the status indicator in the top right corner: a teal dot reading **"live model connected"** confirms the dashboard is successfully communicating with the XGBoost model. If the backend disconnects, an amber **"offline demo mode"** warning will appear, and the system will gracefully fall back to a logistic surrogate.

4. Technical Notes & Limitations

- **Feature Derivation:** The dashboard exposes approximately 11 intuitive controls (e.g., cloud cover, humidity, gust speed). The backend physically derives the remaining variables (such as dew point and vapour pressure deficit) to construct the complete 30+ feature vector expected by the model.
- **Production Scope:** This deployment acts as a faithful interface to the trained model for demonstration and testing purposes. A true production rollout would programmatically pull real-time conditions and compute exact lag features from a live weather API such as Open-Meteo.
- **Threshold Configuration:** The default deploy/stow decision threshold is set at `0.50`. In a production environment, this operating point should be tuned based on the relative business cost of a missed wetting versus an unnecessary deployment.